Summary in Graph

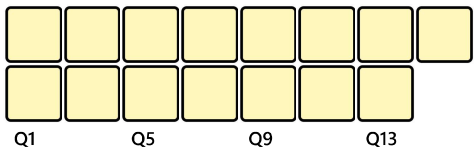
Exam Summary (GO Classes DA Test Series 2026 | Artificial Intelligence | Topic Wise Test 2)

Qs. Attempted:	0 (0 + 0)	Correct Marks:	0 (0 + 0)
Correct Attempts:	0 (0 + 0)	Penalty Marks:	0 (0 + 0)
Incorrect Attempts:	0 (0 + 0)	Resultant Marks:	0 (0 + 0)

Total Questions:	15 (5 + 10)
Total Marks:	25 (5 + 20)
Exam Duration:	45 Minutes
Time Taken:	0 Minutes

Individual Question Statistics

Select Metric: Marks Obtained



* Average Stats based on first-attempt stats of Top 10% (49) out of 490 users

EXAM RESPONSE | EXAM STATS | FEEDBACK

Technical

Q #1 | Multiple Select Type | Award: 1 | Penalty: 0 | Artificial Intelligence

Which of the following statments with respect to a Bayes Net is/are False?

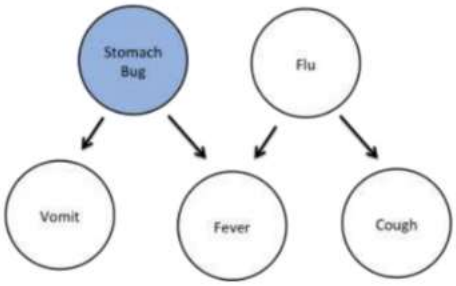
A. Bayes net of variables with no edges (no nodes are connected to any of the others) means all variables are independent of each other.
B. In a Bayesian Network, some Node A is said to be d-separated from B given C if some undirected path between them is blocked with C.

C. A fully-connected Bayesian Network (there is an edge between all possible pair of Nodes in the graph) implies that all variables are independent.
D. Every variable is conditionally dependent on all other variables given its Markov blanket.

Your Answer: | Correct Answer: B;C;D | Not Attempted | Time taken: 00min 08sec | Discuss

Q #2 | Multiple Select Type | Award: 1 | Penalty: 0 | Artificial Intelligence

Consider a Bayes Net with the following graph.



If you do not have a stomach bug, would vomiting affect the probability of having a fever?

A. Yes, vomiting would increase the probability of having a fever.
B. No, vomiting would not change the probability of having a fever.
C. Yes, vomiting would decrease the probability of having a fever.
D. The question cannot be answered with the given information.

Your Answer: | Correct Answer: B | Not Attempted | Time taken: 00min 00sec | Discuss

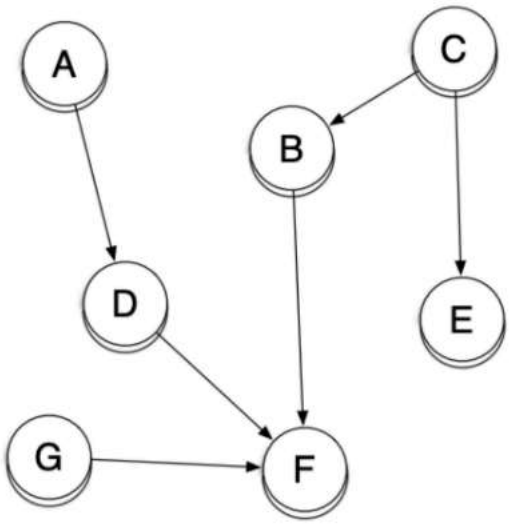
Q #3 | Numerical Type | Award: 1 | Penalty: 0 | Artificial Intelligence

Suppose that you are trying to solve a binary classification problem, and your data set has 3 attributes. Each attribute can take 4 possible values. If you modeled the full joint distribution of the attributes and the class label, how many parameters would you need?

Your Answer: | Correct Answer: 127 | Not Attempted | Time taken: 00min 00sec | Discuss

Q #4 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Artificial Intelligence

Consider a Bayes Net with the following graph.
What is the Markov blanket of node D ?



A. A, D, G, B, F
B. A, E, G, F
C. A, G, F, B, C, E
D. A, F, G, B

Your Answer: | Correct Answer: D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #5 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Artificial Intelligence

Consider five random variables $A, B, C, D,$ and E with the joint distribution factorization:

$$P(A, B, C, D, E) = P(A)P(B \mid A)P(C \mid A, B)P(D \mid B, C)P(E \mid C, D)$$

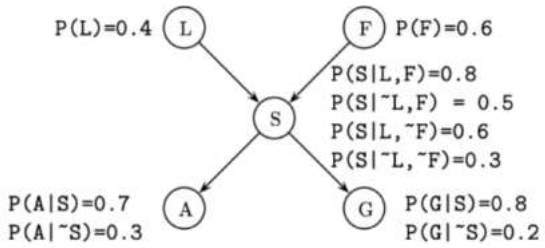
Which statemment is **FALSE?**

A. E is conditionally independent of A given C and D.
B. C is conditionally independent of E given D.
C. B and E are conditionally independent given C and D.
D. D and A are conditionally independent given B and C.

Your Answer: | Correct Answer: B | Not Attempted | Time taken: 00min 00sec | Discuss

Q #6 | Numerical Type | Award: 2 | Penalty: 0 | Artificial Intelligence

A student of the Machine Learning class notices that people driving SUVs (S) consume large amounts of gas (G) and are involved in more accidents than the national average (A). He also noticed that there are two types of people that drive SUVs: people from Pennsylvania (L) and people with large families (F).
After collecting some statistics, he arrives at the following Bayesian network.



What is $P(S)$? (up to two decimal places)

Your Answer: | Correct Answer: 0.54 | Not Attempted | Time taken: 00min 00sec | Discuss

Q #7 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Artificial Intelligence

We are dealing with two probability distributions over N variables, where each variable can take on exactly d values. The distributions are represented by the two Bayes' Nets shown below.



If S is the amount of storage required for the Conditional Probability Tables for $X_2, \ldots, X_N$ in D_1, how much storage is required for the Conditional Probability Tables for $X_2, \ldots, X_N$ in D_2 ?

A. 2^S
B. Sd
C. S^d
D. $S + 2^d$

Your Answer: | Correct Answer: B | Not Attempted | Time taken: 00min 00sec | Discuss

Q #8 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Artificial Intelligence

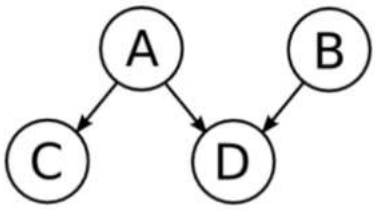
Consider the 3 -stage process: Roll a 6 -sided die (numbered 1-6), call the outcome X. Then, roll an X-sided die (numbered 1-X), call the outcome Y. Finally, roll a Y-sided die (numbered 1- Y), call the outcome Z. What is $P(Z = 5)$?

A. $\frac{61}{5400}$
B. $\frac{89}{5000}$
C. $\frac{91}{5400}$
D. $\frac{93}{5000}$

Your Answer: | Correct Answer: C | Not Attempted | Time taken: 00min 00sec | Discuss

Q #9 | Numerical Type | Award: 2 | Penalty: 0 | Artificial Intelligence

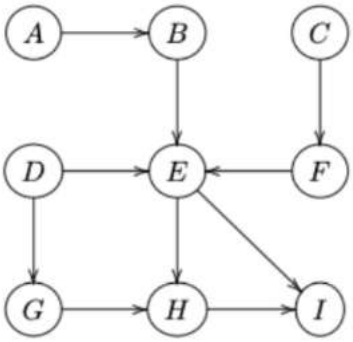
In the following graphical model, $A, B, C,$ and D are binary random variables. How many parameters (minimum) are needed to define the Conditional Probability Distributions (CPD's) for this Bayes Net?



Your Answer: | Correct Answer: 8 | Not Attempted | Time taken: 00min 00sec | Discuss

Q #10 | Multiple Select Type | Award: 2 | Penalty: 0 | Artificial Intelligence

Consider the Bayes' net given below. Remember that $X \perp Y$ means " X is independent of Y given nothing", and $X \perp Y \mid \{Z, W\}$ means " X is independent of Y given Z and W ".



Which of the following statements is/are false?

A. It is guaranteed that $A \perp C$
B. It is guaranteed that $A \perp D \mid E$
C. It is guaranteed that $A \perp I \mid E$
D. It is guaranteed that $B \perp C \mid I$

Your Answer: | Correct Answer: B;C;D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #11 | Multiple Select Type | Award: 2 | Penalty: 0 | Artificial Intelligence

Which of the follwing expressions is/are equivalent to $\mathbf{P}(\mathbf{A}, \mathbf{B} \mid \mathbf{C})$, given that $\mathbf{A} \perp \mathbf{B} \mid \mathbf{C}$? NOTE : $X \perp Y \mid Z$ means X is independent of Y given Z.

A. $\frac{P(C|A)P(A|B)P(B)}{P(C)}$
B. $\frac{P(B,C|A)P(A)}{P(B,C)}$
C. $\frac{P(A|C)P(B,C)}{P(C)}$
D. $P(A \mid B, C)P(B \mid C)$

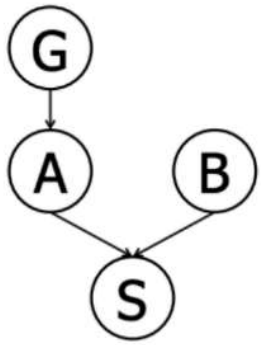
Your Answer: | Correct Answer: C;D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #12 | Numerical Type | Award: 2 | Penalty: 0 | Artificial Intelligence

Suppose that a patient can have a symptom (S) that can be caused by two different diseases (A and B). It is known that the variation of gene G plays a big role in the manifestation of disease A. The Bayes' Net and corresponding conditional probability tables for this situation are shown below.

$\mathbb{P}(G)$	
$+g$	0.1
$-g$	0.9

$\mathbb{P}(A\|G)$		
$+g$	$+a$	1.0
$+g$	$-a$	0.0
$-g$	$+a$	0.1
$-g$	$-a$	0.9



$\mathbb{P}(B)$	
$+b$	0.4
$-b$	0.6

$\mathbb{P}(S\|A,B)$			
$+a$	$+b$	$+s$	1.0
$+a$	$+b$	$-s$	0.0
$+a$	$-b$	$+s$	0.9
$+a$	$-b$	$-s$	0.1
$-a$	$+b$	$+s$	0.8
$-a$	$+b$	$-s$	0.2
$-a$	$-b$	$+s$	0.1
$-a$	$-b$	$-s$	0.9

What is the probability that a patient has the disease carrying gene variation G given that they have disease B ? (rounded off to 1 decimal place).

Your Answer: | Correct Answer: 0.1 | Not Attempted | Time taken: 00min 00sec | Discuss

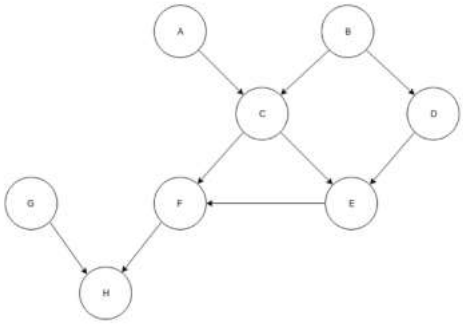
Q #13 | Numerical Type | Award: 2 | Penalty: 0 | Artificial Intelligence

Consider a Naive Bayes classifier with 5 input variables each of which can take 3 possible values and one boolean output, Y. How many minimum number of parameters must be estimated to train such a naive Bayes classifier?

Your Answer: | Correct Answer: 21 | Not Attempted | Time taken: 00min 00sec | Discuss

Q #14 | Multiple Select Type | Award: 2 | Penalty: 0 | Artificial Intelligence

Consider the Bayes' net given below.
Which of the following Statements is/are true?



A. All paths are blocked from H to B.
B. Given C, F; all paths are blocked from H to B.
C. C is conditionally independent of D given F .
D. There exists a node with all other nodes in its Markov Blanket.

Your Answer: | Correct Answer: B | Not Attempted | Time taken: 00min 00sec | Discuss

Q #15 | Multiple Select Type | Award: 2 | Penalty: 0 | Artificial Intelligence

Which of the following statements is/are true with respect to a Bayesian Network?

A. The Bayesian Network with $P(X)P(Y)P(Z \mid X, Y)P(W \mid Y, Z)$ has arcs from X to Z, Y to Z, Y to W, and Z to W, with no other arcs.
B. The product $P(X \mid Y)P(Y \mid Z)P(Z \mid W)P(W \mid X)$ corresponds to a valid Bayesian Network.
C. The Bayesian Network with $P(X)P(Y \mid X)P(Z \mid Y)P(W \mid Z)$ has arcs from X to Y, Y to Z, Z to W, with no other arcs.
D. The product $P(X)P(Y \mid X)P(Z \mid Y, W)P(W \mid X)$ corresponds to a valid Bayesian Network.

Your Answer: | Correct Answer: A;C;D | Not Attempted | Time taken: 00min 00sec | Discuss

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